

Beyond the Level of VRP: Nonlinear Return Predictability in China’s 50ETF

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1 Introduction

This report examines VRP-based return predictability in the Chinese ETF market using the 50ETF (510050.SH). We contribute in two ways. First, we construct VRP using a standard HAR framework for forecasting future realized variance and a VIX-style implied volatility index for extracting implied variance. Second, we evaluate VRP’s incremental explanatory power within a rich predictor space: the Shichuan factor framework, consisting of 54 technical factors widely used in Chinese quantitative finance practice.

Our research questions are:

1. Does VRP alone predict short-horizon (5–20 trading days) forward returns in the Chinese 50ETF market, in a direct replication-style test?
2. Does augmenting a traditional multi-factor technical model with VRP-derived factors increase explanatory power in-sample?

2 Data and Sample

2.1 Data Sources

- **ETF Price Data:** Daily open/high/low/close prices, returns, and trading volume for 50ETF (510050.SH), spanning February 23, 2005 to November 12, 2025 (5,037 observations).
- **Volatility Index:** China VIX 50 index (daily), providing market-implied volatility. The available VIX sample begins on February 9, 2015.

2.2 Effective Sample for VRP and Regressions

VRP construction requires VIX availability and rolling-window features; factor construction further introduces lags and rolling statistics. After merging data and dropping rows with missing values in predictors and targets, the effective regression sample contains:

- **Regression sample size:** 2,271 observations.
- **Date range:** December 8, 2015 to October 15, 2025.

All factors are constructed using only information available up to time t (via shifting and rolling operations), avoiding look-ahead bias in feature definitions.

3 Methodology

3.1 VRP Construction

3.1.1 VRP Summary Statistics (2015–2025).

In the VIX-available sample, VRP exhibits:

- Mean: 0.000156
- Std: 0.001961
- Min/Max: $-0.010617 / 0.012629$
- Sign distribution: 43.5% positive, 56.5% negative

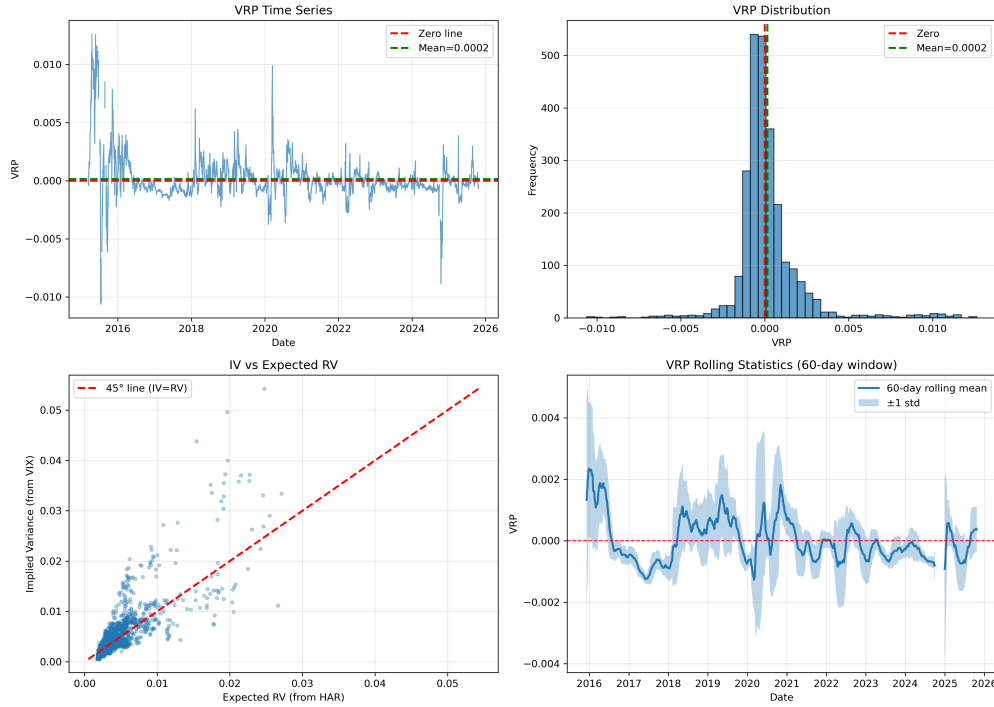


Figure 1: VRP diagnostics

Thus, while the mean VRP is slightly positive, the distribution is centered close to zero with substantial mass on both sides, suggesting that volatility compensation (as captured by this construction) is not persistently positive in the Chinese 50ETF market over the sample.

3.1.2 VRP-Derived Factors

We construct eight VRP-related factors to capture different informational dimensions:

- `vrp`: Raw VRP level.
- `vrp_mom_5d`, `vrp_mom_20d`: VRP momentum (percentage change over 5 and 20 days).
- `vrp_zscore`: Standardized VRP relative to a 60-day rolling mean and standard deviation.
- `vrp_high`, `vrp_low`: Regime indicators based on the 60-day rolling 75th/25th percentiles.
- `vrp_x_rv`: Interaction between VRP and daily realized variance ($VRP_t \cdot RV_t^{(d)}$).

- `iv_rv_spread_alt`: Alternative volatility spread $\sqrt{IV_{t,t+20}} - \sqrt{\widehat{RV}_{t,t+20}^{HAR}}$.

3.2 Shichuan Multi-Factor Framework

We construct 54 technical factors based on the Shichuan factor handbook, grouped into six categories: Momentum Factors (11), Reversal Factors (10), Volatility Factors (12), Liquidity Factors (7), Price Pattern Factors (7), Trend Factors (7). In total, we have **54 Shichuan factors + 8 VRP factors = 62 predictors**.

3.3 Incremental Examination of Individual VRP Components

Apart from combination of 8 VRP factors, it remains unclear which specific component drives this improvement. To address this question, we conduct an incremental examination that decomposes the contribution of VRP at the component level. Specifically, we take the 54-factor Shichuan model as a strong baseline and add *one* VRP-derived factor at a time. For each VRP component $Z_{k,t}$, we estimate:

$$\text{ret}_{t,t+h} = \alpha + \sum_{j=1}^{54} \beta_j \cdot \text{Shichuan}_{j,t} + \gamma_k \cdot Z_{k,t} + \varepsilon_t,$$

where $h \in \{5, 10, 20\}$ denotes the return horizon.

3.4 Target Variables

We study short-horizon forward returns:

$$\begin{aligned} \text{ret_5d_fwd} &= \sum_{k=1}^5 r_{t+k}, \\ \text{ret_10d_fwd} &= \sum_{k=1}^{10} r_{t+k}, \\ \text{ret_20d_fwd} &= \sum_{k=1}^{20} r_{t+k}. \end{aligned}$$

The 20-day horizon aligns with VRP’s construction horizon, while 5–10 day horizons test whether VRP contains shorter-term predictive content.

3.5 Regression Models

We compare four model classes (all evaluated *in-sample* on the effective regression sample).

3.5.1 Model A: VRP-Only (Replication-Style Baseline)

To mirror the single-factor approach, we estimate:

$$\text{ret}_{t,t+h} = \alpha + \beta \cdot \text{VRP}_t + \varepsilon_t,$$

using OLS with Newey–West (HAC) standard errors ($\text{maxlags} = 20$).

3.5.2 Model B: VRP-8 (All VRP-Derived Factors)

We fit a ridge regression using the eight VRP-based predictors to mitigate multicollinearity:

$$\text{ret}_{t,t+h} = \alpha + \sum_{j=1}^8 \beta_j \cdot \text{VRPFactor}_{j,t} + \varepsilon_t,$$

with L_2 penalty parameter $\lambda = 1.0$.

3.5.3 Model C: Shichuan-Only (54 Technical Factors)

We fit ridge regression using the 54 Shichuan factors.

3.5.4 Model D: Full Model (62 Factors)

We fit ridge regression using all 62 factors (54 Shichuan + 8 VRP), and evaluate incremental explanatory power relative to Model C.

3.6 Evaluation Metrics

Since this report focuses on the current in-sample run, we report:

- **In-Sample R^2 and Adjusted R^2** for each model and horizon.
- **VRP coefficient and HAC t-statistics** for the VRP-only specification (Model A).

4 Results

4.1 Main In-Sample Model Comparison

Table 1 summarizes in-sample R^2 across models. The VRP-only model has near-zero R^2 at all horizons. Introducing VRP transformations (VRP-8) increases R^2 to 0.48%, 0.83%, and 1.45% for 5/10/20 days, respectively. The Shichuan factor model explains a substantially larger fraction of variation (5.21%, 7.91%, 13.03%), and adding VRP-8 on top of Shichuan yields a consistent incremental gain, raising R^2 to 5.82%, 8.48%, and 13.54%.

Table 1: In-Sample R^2 Comparison Across Models

Target	VRP-Only (1)	VRP-8 (8)	Shichuan (54)	Full (62)
5-Day Return	0.01%	0.48%	5.21%	5.82%
10-Day Return	0.02%	0.83%	7.91%	8.48%
20-Day Return	0.07%	1.45%	13.03%	13.54%

4.2 VRP-Only: Coefficient and Significance (HAC)

The VRP-only regression (Model A) yields negative but statistically insignificant coefficients at all horizons:

- 5-day: $\hat{\beta} = -0.2657$, $t = -0.28$
- 10-day: $\hat{\beta} = -0.4548$, $t = -0.32$
- 20-day: $\hat{\beta} = -1.1450$, $t = -0.45$

Thus, a direct replication-style test finds no meaningful return predictability from the raw VRP level in this market.

4.3 Incremental Value of VRP Transformations

Although raw VRP has negligible explanatory power, VRP transformations meaningfully improve fit:

- Relative to VRP-only, VRP-8 increases in-sample R^2 by +0.47% (5-day), +0.81% (10-day), and +1.39% (20-day).
- Relative to Shichuan-only, adding VRP-8 increases in-sample R^2 by +0.61% (5-day), +0.58% (10-day), and +0.51% (20-day).

These gains persist even after controlling for a broad set of technical and volatility predictors, suggesting that VRP transformations capture a distinct dimension of market state information.

4.4 Visualization

Figure 2 visualizes model comparisons, highlighting: (i) the near-zero explanatory power of VRP-only; (ii) the improvement from VRP transformations; and (iii) the consistent incremental contribution of VRP factors on top of Shichuan predictors.



Figure 2: In-Sample Model Comparison: R^2 , VRP Transformations Effect, and Incremental Value

4.5 Decomposing the Incremental Contribution of VRP

Table 2: Incremental In-Sample R^2 and Adjusted R^2 from Adding Individual VRP Components (%)

VRP Component	5-Day Return		10-Day Return		20-Day Return	
	ΔR^2	$\Delta \text{Adj. } R^2$	ΔR^2	$\Delta \text{Adj. } R^2$	ΔR^2	$\Delta \text{Adj. } R^2$
Raw VRP (<code>vrp</code>)	+0.038%	−0.005%	−0.000%	−0.043%	+0.051%	+0.012%
VRP Momentum 5d (<code>vrp_mom_5d</code>)	+0.004%	−0.040%	+0.018%	−0.024%	+0.022%	−0.018%
VRP Momentum 20d (<code>vrp_mom_20d</code>)	+0.020%	−0.024%	+0.043%	+0.002%	+0.099%	+0.061%
VRP Z-score (<code>vrp_zscore</code>)	+0.013%	−0.030%	+0.022%	−0.020%	+0.026%	−0.013%
High VRP Regime (<code>vrp_high</code>)	+0.039%	−0.004%	+0.029%	−0.013%	+0.000%	−0.040%
Low VRP Regime (<code>vrp_low</code>)	+0.031%	−0.012%	+0.139%	+0.100%	+0.034%	−0.006%
VRP × RV (<code>vrp_x_rv</code>)	+0.204%	+0.165%	+0.159%	+0.120%	+0.005%	−0.035%
Alt. Vol Spread (<code>iv_rv_spread_alt</code>)	+0.144%	+0.103%	+0.021%	−0.021%	+0.014%	−0.026%

5 Key Findings

This section summarizes the main empirical findings from the in-sample analysis.

5.1 VRP-8 Transformations Improve Model Fit

Although VRP alone has negligible explanatory power, augmenting the model with VRP-derived transformations improves in-sample performance. Relative to the VRP-only specification, the VRP-8 model increases in-sample R^2 by 0.47%, 0.81%, and 1.39% for 5-, 10-, and 20-day forward returns, respectively. Moreover, when added on top of the 54-factor Shichuan baseline, VRP-8 delivers consistent incremental gains of 0.61%, 0.58%, and 0.51% across the same horizons.

5.2 Raw VRP Has Weak Predictive Power in the Chinese Market

A replication-style single-factor regression shows that the raw VRP level has essentially no predictive power for short-horizon returns. Across all horizons, the estimated VRP coefficients are negative and statistically insignificant under Newey–West inference. In-sample R^2 remains close to zero (below 0.1%), in stark contrast to the strong predictive power documented in developed markets.

5.3 Nonlinear VRP Components Drive the Incremental Gains

Decomposing the VRP contribution at the component level reveals that the incremental explanatory power is highly concentrated in a small subset of nonlinear terms. In particular, the interaction term between VRP and realized variance (`vrp_x_rv`) delivers the largest improvements, increasing in-sample R^2 by +0.20% (5-day) and +0.16% (10-day), with corresponding gains in adjusted R^2 .

In addition, regime-based measures such as the low-VRP indicator (`vrp_low`) improve the 10-day horizon by +0.14% in R^2 , while linear or mildly nonlinear transformations (raw VRP, z-scores, simple momentum) contribute little once model complexity is accounted for.

These findings suggest that VRP’s predictive content is fundamentally nonlinear and state-dependent. Volatility mispricing affects returns primarily when conditioned on market risk levels or specific volatility regimes, rather than through the unconditional VRP level itself.

6 Future Research Directions

6.1 Cross-Sectional Tests

A natural extension of this study is to examine whether VRP has cross-sectional predictive power at the individual stock level. Rather than forecasting index returns, one can treat VRP as a market-wide state variable and test its interaction with stock-specific exposures.

6.2 Nonlinear and State-Dependent VRP

The empirical results in this paper indicate that VRP enters return predictability primarily through nonlinear and interaction terms rather than through its raw level. Future work can formalize this insight by modeling VRP as a state-dependent variable whose effect varies with market conditions.

One approach is to explicitly incorporate interaction terms between VRP and realized variance, volatility regimes, or market stress indicators, allowing the marginal effect of VRP to depend on the prevailing risk environment. Alternatively, nonlinear specifications such as threshold models or piecewise-linear regressions can be used to distinguish low- and high-volatility regimes.

From a modeling perspective, these approaches correspond to relaxing the assumption of a constant linear pricing kernel and allowing higher-order or conditional components to influence expected returns. Such frameworks may better capture how volatility mispricing is priced in markets where risk compensation is intermittent rather than constant.

References

- Shichuan. (2019). *Factor Investing: Methods and Practice*. Publishing House of Electronics Industry.
- Bollerslev, T., Tauchen, G., & Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11), 4463–4492.